

Chemical composition of glyphosate-tolerant soybean 40-3-2 grown in Europe remains equivalent with that of conventional soybean (Glycine max L.).

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The composition of glyphosate-tolerant (Roundup Ready) soybean 40-3-2 was compared with that of conventional soybean grown in Romania in 2005 as part of a comparative safety assessment program. Samples were collected from replicated field trials, and compositional analyses were performed to measure proximates (moisture, fat, ash, protein, and carbohydrates by calculation), fiber, amino acids, fatty acids, isoflavones, raffinose, stachyose, phytic acid, trypsin inhibitor, and lectin in grain as well as proximates and fiber in forage. The mean values for all biochemical components assessed for Roundup Ready soybean 40-30-2 were similar to those of the conventional control and were within the published range observed for commercial soybean. The compositional profile of Roundup Ready soybean 40-3-2 was also compared to that of conventional soybean varieties grown in Romania by calculating a 99% tolerance interval to describe compositional variability in the population of traditional soybean varieties already on the marketplace. These comparisons, together with the history of the safe use of soybean as a common component of animal feed and human food, lead to the conclusion that Roundup Ready soybean 40-3-2 is compositionally equivalent to and as safe and nutritious as conventional soybean varieties grown commercially.